

Characterization of Mg^{2+} Inhibition of Mitochondrial Ca^{2+} Uptake by a Mechanistic Model of Mitochondrial Ca^{2+} Uniporter

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ABSTRACT Ca^{2+} is an important regulatory ion and alteration of mitochondrial Ca^{2+} homeostasis can lead to cellular dysfunction and apoptosis. Ca^{2+} is transported into respiring mitochondria via the Ca^{2+} uniporter, which is known to be inhibited by Mg^{2+} . This uniporter-mediated mitochondrial Ca^{2+} transport is also shown to be influenced by inorganic phosphate (Pi). Despite a large number of experimental studies, the kinetic mechanisms associated with the Mg^{2+} inhibition and Pi regulation of the uniporter function are not well established. To gain a quantitative understanding of the effects of Mg^{2+} and Pi on the uniporter function, we developed here a mathematical model based on known kinetic properties of the uniporter and presumed Mg^{2+} inhibition and Pi regulation mechanisms. The model is extended from our previous model of the uniporter that is based on a multistate catalytic binding and interconversion mechanism and Eyring's free energy barrier theory for interconversion. The model satisfactorily describes a wide variety of experimental data sets on the kinetics of mitochondrial Ca^{2+} uptake. The model also appropriately depicts the inhibitory effect of Mg^{2+} on the uniporter function, in which Ca^{2+} uptake is hyperbolic in the absence of Mg^{2+} and sigmoid in the presence of Mg^{2+} . The model suggests a mixed-type inhibition mechanism for Mg^{2+} inhibition of the uniporter function. This model is critical for building mechanistic models of mitochondrial bioenergetics and Ca^{2+} handling to understand the mechanisms by which Ca^{2+} mediates signaling pathways and modulates energy metabolism.

INTRODUCTION

Ca^{2+} is a key regulatory ion and has multiple roles in mitochondrial function and dysfunction (1–5). Ca^{2+} is known to mediate signaling pathways between cytosol and mitochondria and modulate mitochondrial energy metabolism. Alteration of mitochondrial Ca^{2+} homeostasis can lead to mitochondrial dysfunction and mitochondria-mediated cellular injury and apoptosis. Mg^{2+} is another key regulatory ion and the intracellular concentration of free Mg^{2+} varies significantly in vivo (0.2–2 mM) under various physiological and pathological conditions (e.g., hormonal stimulation, myocardial ischemia) (6–8). Alteration of cytosolic Mg^{2+} homeostasis is known to influence mitochondrial substrate and cation (e.g., Ca^{2+}) transport systems and bioenergetics (e.g., Mg^{2+} is a cofactor for the operation of F_1F_0 -ATPase and adenine nucleotide translocase; Mg^{2+} regulates many key enzymes in mitochondria, such as pyruvate dehydrogenase; Mg^{2+} also inhibits mitochondrial Ca^{2+} uptake via the Ca^{2+} uniporter (6–8)). Therefore, it is essential to mechanistically characterize the biophysical and kinetic mechanisms associated with the transport of Ca^{2+} into mitochondria and how variations in cytosolic Mg^{2+} influence this Ca^{2+} transport.

Ca^{2+} uniporter is the primary influx pathway for Ca^{2+} in respiring mitochondria, and hence is an important regulator

of mitochondrial Ca^{2+} . This uniporter-mediated mitochondrial Ca^{2+} uptake is known to be influenced by inorganic phosphate (Pi), in addition to the inhibition by Mg^{2+} (9–13). Thus, the kinetics of Ca^{2+} uptake not only depends on the catalytic properties of the uniporter and the electrochemical gradient of Ca^{2+} across the inner mitochondrial membrane (IMM), but also on the mechanisms of Mg^{2+} inhibition and Pi regulation of the uniporter function. This has been extensively investigated in many experimental (8,9,11–16) and modeling (17–22) studies. Early initial velocity studies of Ca^{2+} uptake in rat heart and liver mitochondria suggest that Mg^{2+} is a competitive inhibitor of Ca^{2+} uptake where it induces a sigmoid kinetics and the degree of sigmoidicity (K_m of Ca^{2+} for the uniporter) increases with increase in $[\text{Mg}^{2+}]$, without altering the maximal activity of the uniporter (9,14,15).

However, subsequent study on the kinetics of Ca^{2+} uptake in rat liver mitochondria reported a noncompetitive type of Mg^{2+} inhibition of Ca^{2+} uptake by increasing the K_m of Ca^{2+} and decreasing the maximal activity of the uniporter, with increase in $[\text{Mg}^{2+}]$ (11). Furthermore, the Ca^{2+} uptake is shown to be regulated by Pi (9,10). Specifically, the kinetic study of Crompton et al. (9) on Ca^{2+} uptake in rat heart mitochondria shows significant differences in the initial Ca^{2+} uptake, when measured with 0 and 1 mM Pi in the buffer. Though the regulatory roles of Pi on oxidative phosphorylation (23,24) and matrix Ca^{2+} buffering through Ca^{2+} -Pi complex formation (25) are well recognized, it is not understood how Pi can modulate Ca^{2+} uptake via the uniporter. Furthermore, though the mechanisms of extramatrix Ca^{2+} and membrane potential

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($\Delta\Psi$) dependencies of Ca^{2+} uptake have been well-described mathematically (17–22), the kinetic mechanism by which Mg^{2+} inhibits and Pi regulates the uniporter function has not been well characterized in terms of a mechanistic model. The need for such a model is critical for developing mechanistic integrated models of mitochondrial bioenergetics and Ca^{2+} handling that can be helpful in understanding the mechanisms by which Ca^{2+} plays a role in mediating signaling pathways and modulating energy metabolism under normal and pathological conditions.

Previously, we introduced a mechanistic mathematical model of Ca^{2+} uniporter (20) that satisfactorily describes the available data on the kinetics of Ca^{2+} uptake, measured in respiring mitochondria isolated from rat hearts and rat livers under diverse experimental preparations (14–16). This model was developed based on a multistate catalytic binding and interconversion mechanism and Eyring's free-energy barrier theory for interconversion (26–28). In addition, the model accounts for the allosteric cooperative binding of Ca^{2+} to the uniporter, observed experimentally (14,15). In a recent work, this model is further improved to accurately characterize the mechanism of $\Delta\Psi$ -dependency of Ca^{2+} uptake (22). An interesting prediction of this improved model is that the Ca^{2+} -binding sites on the uniporter are located at equal and negligible distances from the bulk phase on either side of the IMM and the binding affinities of the uniporter to both cytosolic and matrix Ca^{2+} are the same. This alternate model is shown to be more robust with minimal number of model parameters that accurately describes the available data (14–16). Furthermore, this alternate model is insensitive to physiological variations in matrix $[\text{Ca}^{2+}]$. Though these models are able to characterize the possible mechanisms of both the extramatrix Ca^{2+} and $\Delta\Psi$ dependencies of Ca^{2+} uptake, they do not account for the effects of Mg^{2+} and Pi on the kinetics of Ca^{2+} uptake. Furthermore, there exists no mathematical model in the literature that characterizes the inhibitory effect of Mg^{2+} and regulatory role of Pi on the uniporter-mediated Ca^{2+} uptake in mitochondria.

In this article, we develop a mechanistic mathematical model of Ca^{2+} uniporter that improves the existing models (17–22) and provides a biophysical basis for the inhibitory effect of Mg^{2+} and regulatory role of Pi on the uniporter-mediated Ca^{2+} uptake. The model, extended from our previous models (20,22), is thermodynamically balanced and adequately describes a large number of independent data sets on both Mg^{2+} -dependent and Mg^{2+} -independent Ca^{2+} uptake via the uniporter (8,9,11,12,14–16). The model accurately depicts the inhibitory effect of Mg^{2+} on Ca^{2+} uptake in which the kinetics of Ca^{2+} uptake is hyperbolic in the absence of Mg^{2+} and sigmoid in the presence of Mg^{2+} . A mixed type Mg^{2+} inhibition mechanism is shown to be consistent with the available data.

MATHEMATICAL FORMULATION

Experimental data for model formulation

The kinetic mechanisms associated with the Mg^{2+} inhibition and Pi regulation of mitochondrial Ca^{2+} transport is not well established. However, experimental studies in respiring isolated mitochondria (9,11,14–16) and intact permeabilized myocytes (8) provide adequate information about the influences of Mg^{2+} and Pi on the initial rates of Ca^{2+} transport into mitochondria. The initial rate data from isolated mitochondria describe the kinetics of extramatrix Ca^{2+} - and $\Delta\Psi$ -dependent mitochondrial Ca^{2+} uptake at various extramatrix $[\text{Mg}^{2+}]$ and $[\text{Pi}]$. The initial rate data from permeabilized cells demonstrate the nature of mitochondrial Ca^{2+} uptake in response to physiological variations of cytosolic $[\text{Mg}^{2+}]$ and $[\text{Ca}^{2+}]$. Therefore, these kinetic data are appropriate for building kinetic models of the Ca^{2+} uniporter to characterize the mechanisms by which Mg^{2+} and Pi regulate the uniporter-mediated mitochondrial Ca^{2+} transport.

The kinetic studies of Scarpa and colleagues (14,15) report the initial rates of Ca^{2+} uptake with variations in extramatrix $[\text{Ca}^{2+}]$ (0–200 μM), measured in isolated mitochondria from rat livers and rat hearts, in the presence of 2 and 5 mM Mg^{2+} and 5 and 2 mM Pi in the buffer. These studies reveal a sigmoid kinetics of the extramatrix Ca^{2+} -dependent Ca^{2+} uptake in the presence of Mg^{2+} and Pi. The kinetic study of Crompton et al. (9) reports the initial rates of Ca^{2+} uptake with variations in extramatrix $[\text{Ca}^{2+}]$ (0–35 μM), measured in isolated mitochondria from rat hearts, both in the absence and presence of 1 mM Mg^{2+} and Pi in the buffer. Although this study reveals a similar sigmoid kinetics of the extramatrix Ca^{2+} -dependent Ca^{2+} uptake in the presence of 1 mM Mg^{2+} and 1 mM Pi, the kinetics was shown to become hyperbolic in the absence of Mg^{2+} with 0 and 1 mM Pi present in the buffer. These studies suggest that Mg^{2+} is a competitive inhibitor of Ca^{2+} uptake that alters only the K_m of Ca^{2+} for the uniporter. The kinetic study of Bragadin et al. (11) reports the initial rates of Ca^{2+} uptake with variations in extramatrix $[\text{Ca}^{2+}]$ (0–300 μM), measured in isolated mitochondria from rat livers, both in the absence and presence of 2 and 5 mM Mg^{2+} with 2 mM Pi in the buffer. This study reveals a similar sigmoid kinetics of the extramatrix Ca^{2+} -dependent Ca^{2+} uptake in the presence of 2 and 5 mM Mg^{2+} , and the kinetic was shown to become hyperbolic in the absence of Mg^{2+} .

However, this study indicates a noncompetitive type of Mg^{2+} inhibition of Ca^{2+} uptake that significantly alters both the K_m of Ca^{2+} and the maximal activity of the uniporter. The kinetic study of Wingrove et al. (16) reports the initial rates of Ca^{2+} uptake with variations in $\Delta\Psi$ (90–200 mV) for different levels of extramatrix Ca^{2+} ($[\text{Ca}^{2+}]_e = 0.5, 1.0, \text{ and } 1.5 \mu\text{M}$), measured in isolated mitochondria from rat livers, in the absence of Mg^{2+} with

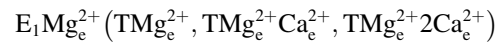
0.2 mM Pi present in the buffer. This study reveals a nonlinear, Goldman-Hodgkin-Katz-type $\Delta\Psi$ -dependency of Ca²⁺ uptake. The regulatory effect of Pi on Ca²⁺ uptake seems to be prominent only at low Pi levels (0–1 mM), which is evident from the study of Crompton et al. (9): Pi was shown to increase both the K_m of Ca²⁺ and the maximal activity of the uniporter; Pi did not appreciably affect the kinetics of Ca²⁺ uptake at higher levels of Pi ([Pi] > 1 mM). This kinetic model of Ca²⁺ uniporter is developed based on basic physical-chemical principles of ion transport across biomembranes and is parameterized to adequately reproduce these diverse experimental data sets in isolated mitochondria as well as the data sets of Szanda et al. (8) in permeabilized cells.

Proposed mechanism of Mg²⁺ inhibition of Ca²⁺ uniporter

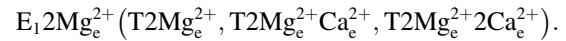
Fig. 1 schematizes the proposed kinetic mechanism of Mg²⁺ inhibition of the uniporter-mediated Ca²⁺ transport into mitochondria. The uniporter functional unit is assumed to have two binding sites for Ca²⁺ and two distinct allosteric sites for Mg²⁺, facing either the external (cytosolic) side or the internal (matrix) side depending on the state of the uniporter (Fig. 1 A). In the forward process, an ionized

Ca²⁺ from the cytosolic side binds to the uniporter T to form the complex TCa²⁺_e, which further favors the binding of second ionized Ca²⁺ (cooperative binding) to form the complex T2Ca²⁺_e. The complex T2Ca²⁺_e then undergoes conformational changes forming the complex 2Ca²⁺_xT in the matrix side. In the reverse process, the complex 2Ca²⁺_xT dissociates in two steps to form two free Ca²⁺ ions and the unbound uniporter T in the matrix side (Fig. 1 B).

We consider a general, linear mixed-type inhibition mechanism for Mg²⁺ inhibition of the uniporter function. In this scheme, the interactions of Mg²⁺ and Ca²⁺ with the uniporter in the cytosolic side can be described by Fig. 1 C. Here, in one process, two Mg²⁺ ions cooperatively bind to the uniporter states E₁ (T, TCa²⁺_e, T2Ca²⁺_e) in two different steps to form the complexes



and



In another process, two Ca²⁺ ions cooperatively bind to the uniporter states E₂ (T, TMg²⁺_e, T2Mg²⁺_e) in two different steps to form the complexes

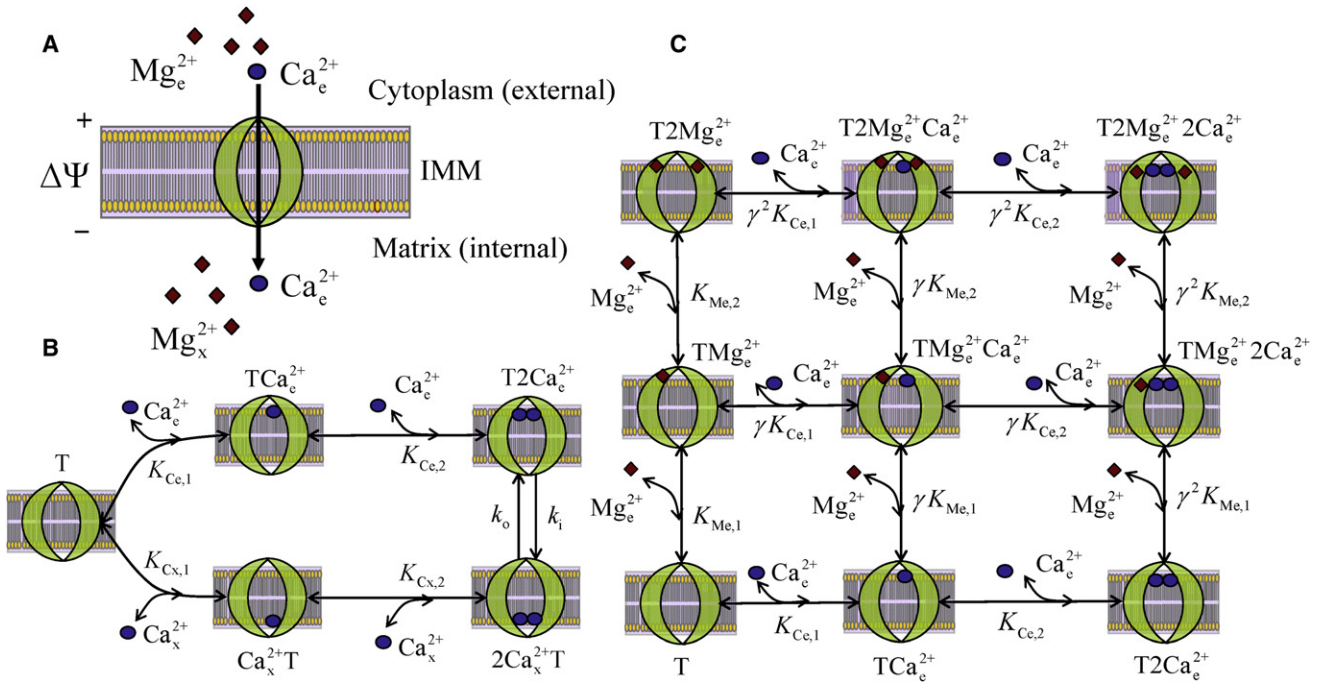


FIGURE 1 Proposed kinetic mechanism of Mg²⁺ inhibition of mitochondrial Ca²⁺ uptake via the Ca²⁺ uniporter. (A) Schematics of the Ca²⁺ uniporter, driven by the electrochemical gradient of Ca²⁺ (circle) across the IMM, exposed to Mg²⁺ ions (diamond). (B) Schematics of different Ca²⁺-bound uniporter states during translocation of Ca²⁺ across the IMM. (C) Schematics of different Ca²⁺- and Mg²⁺-bound uniporter states in the cytoplasmic side in a linear mixed-type inhibition mechanism for Mg²⁺ inhibition of mitochondrial Ca²⁺ uptake. In the matrix side, the Ca²⁺-Mg²⁺ interaction is symmetrical to that in the cytoplasmic side. $K_{Ce,1}$, $K_{Ce,2}$, $K_{Me,1}$, and $K_{Me,2}$ are the dissociation constants associated with the binding of first and second cytoplasmic Ca²⁺ and Mg²⁺ to the uniporter; $1 \leq \gamma \leq \infty$ is a parameter that controls the binding affinity of Ca²⁺ and Mg²⁺ to the uniporter. The translocation of Ca²⁺ across the IMM (T2Ca²⁺_e ↔ 2Ca²⁺_x) is limited by two rate constants k_i and k_o , which are dependent on $\Delta\Psi$.

$$E_2Ca_e^{2+}(TCa_e^{2+}, TMg_e^{2+}Ca_e^{2+}, T_2Mg_e^{2+}Ca_e^{2+})$$

and

$$E_22Ca_e^{2+}(T_2Ca_e^{2+}, TMg_e^{2+}2Ca_e^{2+}, T_2Mg_e^{2+}2Ca_e^{2+}).$$

In the matrix side, the Ca^{2+} - Mg^{2+} interaction is symmetrical to that in the cytoplasmic side. The complete kinetic scheme for the Mg^{2+} inhibition of the uniporter function is shown in Fig. S1 of Appendix SA in the Supporting Material. Except for the $2Ca^{2+}$ -bound uniporter complexes $T_2Ca_e^{2+}$ and $2Ca_x^{2+}T$, the other uniporter complexes are assumed not to undergo any conformational changes that give rise to Ca^{2+} translocation across the IMM.

The translocation of Ca^{2+} via the uniporter is limited by the interconversion rate constants k_i and k_o , which are dependent on $\Delta\Psi$. K_{Ce} and K_{Cx} (K_{Me} and K_{Mx}) are the dissociation constants for the binding of cytosolic and matrix Ca^{2+} (Mg^{2+}) to the uniporter. It is assumed that the binding of Mg^{2+} affects the binding of Ca^{2+} and vice versa. In this regard, the binding affinities of Ca^{2+} to the Mg^{2+} -bound uniporter states and Mg^{2+} to the Ca^{2+} -bound uniporter states decrease with increase in the number of Ca^{2+} and Mg^{2+} binding, which is mainly determined by the control parameter γ ($1 \leq \gamma \leq \infty$). Note that the parameter γ defines the type of Mg^{2+} inhibition of the uniporter function. For example, $\gamma = 1$ represents a noncompetitive inhibition, $\gamma = \infty$ represents a competitive inhibition, and $1 < \gamma < \infty$ represents a mixed-type inhibition mechanism.

Ca^{2+} uniporter flux expression

Based on the proposed mechanism of Mg^{2+} inhibition of mitochondrial Ca^{2+} transport (Fig. 1) and the assumption of quasi-steady-state flux and rapid equilibrium binding of Ca^{2+} and Mg^{2+} , the net Ca^{2+} transport flux via the uniporter (J_{CU}) can be expressed as (see Appendix SA in the Supporting Material):

$$J_{CU} = \frac{[T]_{tot}}{D} \left(k_i \frac{[Ca^{2+}]_e^2}{K_{Ce}^2} - k_o \frac{[Ca^{2+}]_x^2}{K_{Cx}^2} \right), \quad (1)$$

where $[T]_{tot}$ represents the total uniporter concentration, and

Model 1

$$D = D_1 = 1 + \frac{[Ca^{2+}]_e^2}{K_{Ce}^2} + \frac{[Ca^{2+}]_x^2}{K_{Cx}^2} + \frac{[Mg^{2+}]_e^2}{K_{Me}^2} + \frac{[Mg^{2+}]_x^2}{K_{Mx}^2} + \frac{[Ca^{2+}]_e^2[Mg^{2+}]_e^2}{\gamma^4 K_{Ce}^2 \cdot K_{Me}^2} + \frac{[Ca^{2+}]_x^2[Mg^{2+}]_x^2}{\gamma^4 K_{Cx}^2 \cdot K_{Mx}^2}, \quad (2a)$$

Model 2

$$D = D_2 = 1 + \frac{[Ca^{2+}]_e}{K_{Ce}} + \frac{[Ca^{2+}]_e^2}{K_{Ce}^2} + \frac{[Ca^{2+}]_x}{K_{Cx}} + \frac{[Ca^{2+}]_x^2}{K_{Cx}^2} + \frac{[Mg^{2+}]_e}{K_{Me}} + \frac{[Mg^{2+}]_e^2}{K_{Me}^2} + \frac{[Mg^{2+}]_x}{K_{Mx}} + \frac{[Mg^{2+}]_x^2}{K_{Mx}^2} + \frac{[Ca^{2+}]_e[Mg^{2+}]_e}{\gamma \cdot K_{Ce} \cdot K_{Me}} + \frac{[Ca^{2+}]_e[Mg^{2+}]_e^2}{\gamma^2 \cdot K_{Ce} \cdot K_{Me}^2} + \frac{[Ca^{2+}]_e^2[Mg^{2+}]_e}{\gamma^2 \cdot K_{Ce}^2 \cdot K_{Me}} + \frac{[Ca^{2+}]_e^2[Mg^{2+}]_e^2}{\gamma^4 \cdot K_{Ce}^2 \cdot K_{Me}^2} + \frac{[Ca^{2+}]_x[Mg^{2+}]_x}{\gamma \cdot K_{Cx} \cdot K_{Mx}} + \frac{[Ca^{2+}]_x[Mg^{2+}]_x^2}{\gamma^2 \cdot K_{Cx} \cdot K_{Mx}^2} + \frac{[Ca^{2+}]_x^2[Mg^{2+}]_x}{\gamma^2 \cdot K_{Cx}^2 \cdot K_{Mx}} + \frac{[Ca^{2+}]_x^2[Mg^{2+}]_x^2}{\gamma^4 \cdot K_{Cx}^2 \cdot K_{Mx}^2}. \quad (2b)$$

Model 3

$$D = D_3 = 1 + 2 \frac{[Ca^{2+}]_e}{K_{Ce}} + \frac{[Ca^{2+}]_e^2}{K_{Ce}^2} + 2 \frac{[Ca^{2+}]_x}{K_{Cx}} + \frac{[Ca^{2+}]_x^2}{K_{Cx}^2} + 2 \frac{[Mg^{2+}]_e}{K_{Me}} + \frac{[Mg^{2+}]_e^2}{K_{Me}^2} + 2 \frac{[Mg^{2+}]_x}{K_{Mx}} + \frac{[Mg^{2+}]_x^2}{K_{Mx}^2} + 4 \frac{[Ca^{2+}]_e[Mg^{2+}]_e}{\gamma \cdot K_{Ce} \cdot K_{Me}} + 2 \frac{[Ca^{2+}]_e[Mg^{2+}]_e^2}{\gamma^2 \cdot K_{Ce} \cdot K_{Me}^2} + 2 \frac{[Ca^{2+}]_e^2[Mg^{2+}]_e}{\gamma^2 \cdot K_{Ce}^2 \cdot K_{Me}} + \frac{[Ca^{2+}]_e^2[Mg^{2+}]_e^2}{\gamma^4 \cdot K_{Ce}^2 \cdot K_{Me}^2} + 4 \frac{[Ca^{2+}]_x[Mg^{2+}]_x}{\gamma \cdot K_{Cx} \cdot K_{Mx}} + 2 \frac{[Ca^{2+}]_x[Mg^{2+}]_x^2}{\gamma^2 \cdot K_{Cx} \cdot K_{Mx}^2} + 2 \frac{[Ca^{2+}]_x^2[Mg^{2+}]_x}{\gamma^2 \cdot K_{Cx}^2 \cdot K_{Mx}} + \frac{[Ca^{2+}]_x^2[Mg^{2+}]_x^2}{\gamma^4 \cdot K_{Cx}^2 \cdot K_{Mx}^2}. \quad (2c)$$

Note that for $\gamma > 1$, the higher-order terms in the uniporter flux expression becomes smaller and smaller compared to the leading order terms, because the concentrations of Ca^{2+} and Mg^{2+} bound uniporter states become smaller and smaller. Model 1, Model 2, and Model 3 represent, respectively, the full cooperative, partial cooperative, and no-cooperative binding of Ca^{2+} and Mg^{2+} to the uniporter (see Appendix SA in the Supporting Material). The kinetic parameters K_{Ce} , K_{Cx} , k_i , and k_o are constrained by the equilibrium relationship

$$K_{eq} = \frac{k_i \cdot K_{Cx}^2}{k_o \cdot K_{Ce}^2} = \left(\frac{[Ca^{2+}]_x^2}{[Ca^{2+}]_e^2} \right)_{eq}, \quad (3)$$

where K_{eq} is the equilibrium constant for transport of $2Ca^{2+}$ via the uniporter.

$\Delta\Psi$ -dependency of the kinetic parameters

We follow a similar approach to that of our previous cation transporter models (20,22,29) to describe the $\Delta\Psi$ -dependency of the kinetic parameters. Because Mg^{2+} and Ca^{2+} have identical valence, similar assumptions and biophysical parameters are used to describe the $\Delta\Psi$ -dependency of Mg^{2+} and Ca^{2+} binding constants for the uniporter. We only summarize below the final equations that are used to express the kinetic parameters K_{eq} , K_{Ce} , K_{Cx} , K_{Me} , K_{Mx} , k_i , and k_o in terms of $\Delta\Psi$:

$$K_{eq} = \exp(2\Delta\Phi), \quad \Delta\Phi = \frac{ZF\Delta\Psi}{RT}, \quad (4)$$

$$K_{Ce} = K_{Ce}^0 \exp(-\alpha_e \Delta\Phi), \quad K_{Cx} = K_{Cx}^0 \exp(+\alpha_x \Delta\Phi), \quad (5)$$

$$K_{Me} = K_{Me}^0 \exp(-\alpha_e \Delta\Phi), \quad K_{Mx} = K_{Mx}^0 \exp(+\alpha_x \Delta\Phi),$$

$$k_i = k_i^0 \exp(+2\beta_e \Delta\Phi), \quad k_o = k_o^0 \exp(-2\beta_x \Delta\Phi), \quad (6)$$

$$\beta_e = \frac{1}{2} \left[1 + \frac{nH}{\Delta\Phi} \ln \left\{ \frac{\left(\frac{\Delta\Phi}{nH} \right)}{\sinh \left(\frac{\Delta\Phi}{nH} \right)} \right\} \right] - \alpha_e, \quad (7)$$

$$\beta_x = \frac{1}{2} \left[1 - \frac{nH}{\Delta\Phi} \ln \left\{ \frac{\left(\frac{\Delta\Phi}{nH} \right)}{\sinh \left(\frac{\Delta\Phi}{nH} \right)} \right\} \right] - \alpha_x.$$

Here $Z = 2$ is the valence of Ca^{2+} or Mg^{2+} ; F , R , and T denote the Faraday's constant, ideal gas constant, and absolute temperature, respectively ($F = 96484.6$ J/mol/mV and $R = 8.3145$ J/mol/K); α_e (α_x) is the ratio of the potential difference between Ca^{2+} or Mg^{2+} bound to the site of the uniporter facing the external (internal) side of the IMM and Ca^{2+} or Mg^{2+} in the bulk phase to the total membrane potential; and β_e (β_x) is the displacement of external (internal) Ca^{2+} from the coordinate of maximum potential barrier. The $\Delta\Psi$ -dependency of k_i and k_o are derived based on a nonlinear Goldman-Hodgkin-Katz type formalism by expressing β_e and β_x as functions of $\Delta\Psi$ (22); nH is an arbitrary number that determines the degree of deviation from the linear Goldman-Hodgkin-Katz formalism.

Substituting Eq. 4 for K_{eq} , Eq. 5 for K_{Ce} and K_{Cx} , and Eq. 6 for k_i and k_o into Eq. 3, the following kinetic and thermodynamic constraints were obtained:

$$\left(\frac{k_i^0}{k_o^0} \right) \cdot \left(\frac{K_{Cx}^0}{K_{Ce}^0} \right)^2 = 1 \quad \text{and} \quad \alpha_e + \alpha_x + \beta_e + \beta_x = 1. \quad (8)$$

From our previous analyses of the $\Delta\Psi$ -dependency of mitochondrial Ca^{2+} uptake (22), it is apparent that the Ca^{2+} (Mg^{2+}) binding sites on the uniporter are located at equal and negligible distances from the bulk phase on either side of the IMM (i.e., $\alpha_e = \alpha_x \approx 0$) and the binding affinities of the uniporter to both cytosolic and matrix Ca^{2+} (Mg^{2+}) are the same ($K_{Ce}^0 = K_{Cx}^0$ and K_{Me}^0 and K_{Mx}^0). The kinetic constraint in Eq. 8, then, provides $k_i^0 = k_o^0$ and the biophysical constraint in Eq. 8 is automatically satisfied. The estimate $nH = 2.65$ was found to well describe the $\Delta\Psi$ -dependencies of the biophysical parameters β_e and β_x . We followed similar assumptions for binding of Ca^{2+} and Mg^{2+} to the uniporter and translocation of Ca^{2+} via the uniporter, so that $\alpha_e = \alpha_x = \alpha \approx 0$, $K_{Ce}^0 = K_{Cx}^0 = K_C^0$, $K_{Me}^0 = K_{Mx}^0 = K_M^0$, $k_i^0 = k_o^0 = k^0$, and $nH = 2.65$. With this, the number of unknown parameters for estimation in this model is further reduced from ten to four (k^0 , K_C^0 , K_M^0 , and γ).

Pi-dependency of the kinetic parameters

Low extramatrix Pi levels have been shown to modulate the kinetics of Ca^{2+} uniporter in isolated rat heart mitochondria. Specifically, the kinetic study of Crompton et al. (9) shows changes in both the K_m of Ca^{2+} and the maximal activity of the uniporter at 0 and 1 mM Pi, in the presence and absence of extramatrix Mg^{2+} . However, other initial rate studies with higher concentrations of Pi (>1 mM) do not show any significant effect of Pi on the kinetics of Ca^{2+} uniporter (11,14,15). In view of these experimental observations, we incorporate the Pi effect on the uniporter function by exclusively modifying the apparent K_m of Ca^{2+} and Mg^{2+} for the uniporter (K_C^0 and K_M^0) as

$$K_C^0 = K_C^0 \cdot \left(1 + \frac{[Pi]}{(K_{Pi} + [Pi])} \right)$$

and

$$K_M^0 = \frac{K_M^0}{\left(1 + \frac{[Pi]}{(K_{Pi} + [Pi])} \right)}, \quad (9)$$

where K_{Pi} is the Pi binding constant for the uniporter; K_C^0 and K_M^0 on the right side of Eq. 9 are the true (intrinsic) dissociation constants of Ca^{2+} and Mg^{2+} for the uniporter. Further justification for the proposed mechanism of Pi regulation of the kinetic parameters K_C^0 and K_M^0 is given in Appendix SB in the Supporting Material. With negligible pH gradient across IMM, Pi can quickly equilibrate across the IMM via the Pi-H⁺ cotransporter (23,24,30). Therefore, we assumed that $[Pi]_e = [Pi]_x = [Pi]$. With this modification, the total number of unknown parameters for estimation in this uniporter model becomes five (k^0 , K_C^0 , K_M^0 , K_{Pi} and γ).

Parameter estimation and sensitivity analysis

To estimate the unknown kinetic parameters $\phi = (k_0, K_C^0, K_M^0, K_{Pi}, \gamma)$ of the model, a combined least-squares estimation technique was used in multiple steps to fit the model-simulated outputs to the available experimental data,

$$\min_{\phi} E(\phi), \quad E(\phi) = \sum_i^{N_{\text{exp}}} \sum_j^{N_{\text{data}}} \frac{1}{N_{\text{data}}} \left(\frac{J_{\text{CU},j}^{\text{data}} - J_{\text{CU},j}^{\text{model}}(\phi)}{\max(J_{\text{CU},j}^{\text{data}})} \right)^2, \quad (10)$$

where N_{exp} is the number of experiments and N_{data} is the number of data points in a particular experiment, $J_{\text{CU},j}^{\text{data}}$ are the experimental data and $J_{\text{CU},j}^{\text{model}}(\phi)$ are the corresponding model simulated outputs, and $\max(J_{\text{CU},j}^{\text{data}})$ is the maximum value of $J_{\text{CU},j}^{\text{data}}$. The accuracy and robustness of the model fitting to the data for a particular model are assessed based on the value of mean residual error $E(\phi)$ and its sensitivities to perturbations in the optimal parameter estimates.

The model sensitivity to infinitesimal changes in the parameter values can be computed by

$$S_{\phi_i}^E = \left| \frac{\partial \ln E}{\partial \ln \phi_i} \right| = \frac{\phi_i}{E} \left| \frac{\partial E}{\partial \phi_i} \right| \approx \left| \frac{E(\phi_i + 0.001\phi_i) - E(\phi_i - 0.001\phi_i)}{0.002E(\phi_i)} \right|; \quad (11)$$

$$S_{\phi}^E = \sum_i^{N_p} |S_{\phi_i}^E|,$$

where ϕ_i is the i^{th} parameter; $S_{\phi_i}^E$ denotes the normalized sensitivity index for the parameter ϕ_i , whereas S_{ϕ}^E denote the combined normalized sensitivity index for all parameters together. A higher sensitivity index indicates that a small change in a given parameter (or all parameters) can result in a significant change in the model output. The most accurate model and parameters can have higher $S_{\phi_i}^E$ and S_{ϕ}^E with lower least-squared error $E(\phi)$.

RESULTS

In this section, we provide the detailed parameterization and independent validation of the three versions of the kinetic model of mitochondrial Ca^{2+} uniporter. The three kinetic models (Model 1, full cooperativity; Model 2, partial cooperativity; Model 3, no cooperativity) are simulated to exclusively fit the available data sets (9,11,14–16) on the kinetics of extramatrix Ca^{2+} - and $\Delta\Psi$ -dependent mitochondrial Ca^{2+} uptake measured both in the absence and presence of different amounts of Mg^{2+} and Pi in the buffer. The model fittings of these data sets are demonstrated through Fig. 2. The dashed, solid, and dotted lines represent the fittings for Model 1, Model 2, and Model 3, respectively. The model is further validated by comparing the model

predictions to another data set (8) on the kinetics of mitochondrial Ca^{2+} uptake measured in permeabilized cells with physiological variations in cytosolic $[\text{Ca}^{2+}]$ and $[\text{Mg}^{2+}]$, which is demonstrated in Fig. 3. The estimated parameter values associated with these three variant models are provided in Table 1. Additional model simulations describing the nature of extramatrix Ca^{2+} and $\Delta\Psi$ dependencies and the nature of Mg^{2+} inhibition of mitochondrial Ca^{2+} uptake are demonstrated through Fig. S3, Fig. S4, and Fig. S5 of Appendix SC in the Supporting Material.

We followed a multistep modular approach for parameterization of the model:

- Step 1. The kinetic parameters k^0 and K_C^0 were estimated based on the Pi- and Mg^{2+} -independent data ($[\text{Pi}]_e = 0$ mM; $[\text{Mg}^{2+}]_e = 0$ mM) of Crompton et al. (9) (Fig. 2 A). The values of the kinetic parameters K_{Pi} , K_M^0 , and γ were reasonably chosen (fixed) to fit this data set, because these parameters are insensitive to this data set. The $\Delta\Psi$ was set at 140 mV, corresponding to 0 mM Pi, based on a 2003 study by Bose et al. (23).
- Step 2. The Pi-dependent data of Crompton et al. (9) (Fig. 2 B) both with and without Mg^{2+} ($[\text{Pi}]_e = 1$ mM; $[\text{Mg}^{2+}]_e = 0$ and 1 mM) were used to estimate K_{Pi} , K_M^0 , and γ , as these data sets are suitable to estimate the Pi and Mg^{2+} binding constants. In this step, the values of k^0 and K_C^0 remained the same, as estimated in the first step. The $\Delta\Psi$ was set at 190 mV, corresponding to a typical state-2 respiration with $\text{Pi} \geq 0.2$ mM (23,24,30).
- Step 3. The five parameters k^0 , K_C^0 , K_M^0 , K_{Pi} and γ were further optimized and fine-tuned to jointly fit the model to the three data sets of Crompton et al. (9), along with the kinetic data of Bragadin et al. (11) ($[\text{Pi}]_e = 2$ mM; $[\text{Mg}^{2+}]_e = 0, 2$, and 5 mM, see our Fig. 2 C); Vinogradov and Scarpa (14) ($[\text{Pi}]_e = 5$ mM; $[\text{Mg}^{2+}]_e = 2$ mM, see our Fig. 2 D); Scarpa and Graziotti (15) ($[\text{Pi}]_e = 2$ mM; $[\text{Mg}^{2+}]_e = 5$ mM, see our Fig. 2 E); and Wingrove et al. (16) ($[\text{Pi}]_e = 0.2$ mM; $[\text{Mg}^{2+}]_e = 0$ mM, see our Fig. 2 F).

The parameter k^0 was allowed to vary among data sets from different groups to account for different tissue types (i.e., mitochondria from rat livers versus rat hearts) and diverse experimental preparations that affect the overall uniporter activity (e.g., heterogeneity of mitochondrial population and differential energization or $\Delta\Psi$). The biophysical parameter $\alpha_e = \alpha_e = \alpha$ was set at zero and nH was set at 2.65, based on our previous uniporter model (22). Matrix $[\text{Ca}^{2+}]$ was set at 250 nM; matrix $[\text{Mg}^{2+}]$ was set at 0.4 mM, a value within the physiological range, based on the extensive work of Jung et al. (31); and $\Delta\Psi$ was set at 190 mV for $\text{Pi} \geq 0.2$ mM (9,23). We note here that, this extended uniporter has all the characteristics of our previous uniporter model (20,22). The model is insensitive to

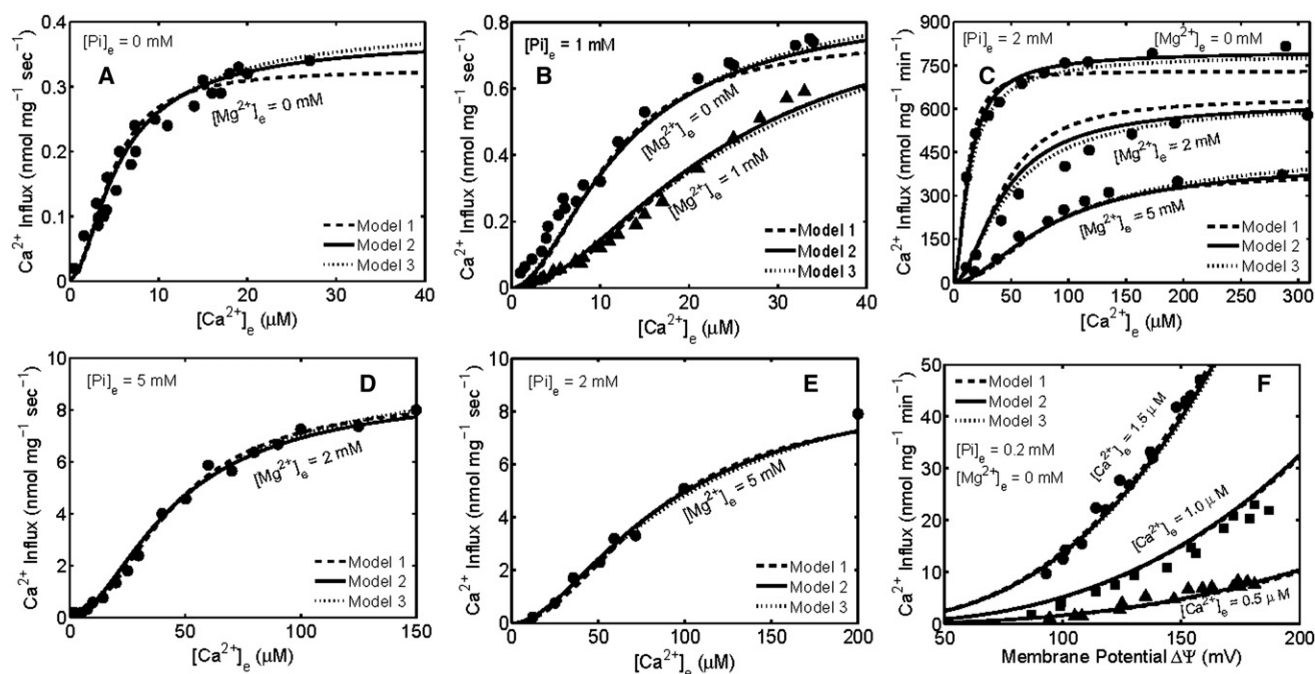
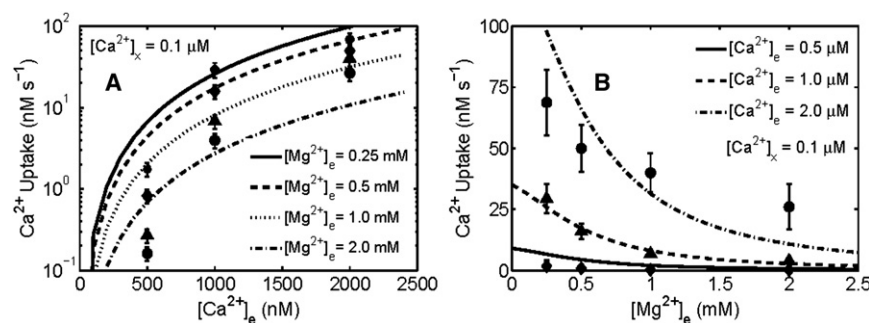


FIGURE 2 Comparison of the Ca²⁺ uniporter models (lines) to the experimental data (points) on the extramatrix Ca²⁺ and $\Delta\Psi$ -dependencies of mitochondrial Ca²⁺ uptake measured both in the presence and absence of extramatrix Mg²⁺. Shown are the fittings of three different versions of the kinetic model (Model 1, Model 2, and Model 3) to the kinetic data of (A and B) Crompton et al. (9) in which the initial rates of Ca²⁺ influx in response to variations in extramatrix [Ca²⁺]_e were measured in energized mitochondria of rat hearts suspended (A) in a medium containing 0 mM Mg²⁺ and 0 mM Pi and (B) in a medium containing 0 and 1 mM Mg²⁺ and 1 mM Pi; (C) Bragadin et al. (11) in which the initial rates of Ca²⁺ influx in response to variations in extramatrix [Ca²⁺]_e were measured in energized mitochondria of rat livers suspended in a medium containing different levels of Mg²⁺ ([Mg²⁺]_e = 0, 2, and 5 mM) and 2 mM Pi; (D) Vinogradov and Scarpa (14) in which the initial rates of Ca²⁺ influx in response to variations in extramatrix [Ca²⁺]_e were measured in energized mitochondria of rat livers suspended in a medium containing 2 mM Mg²⁺ and 5 mM Pi; (E) Scarpa and Graziotti (15) in which the initial rates of Ca²⁺ influx in response to variations in extramatrix [Ca²⁺]_e were measured in energized mitochondria of rat hearts suspended in a medium containing 5 mM Mg²⁺ and 2 mM Pi; and (F) Wingrove et al. (16) in which the initial rates of Ca²⁺ influx as a function of $\Delta\Psi$ were measured in energized mitochondria of rat livers suspended in a medium containing three different levels of Ca²⁺ ([Ca²⁺]_e = 0.5, 1.0, and 1.5 μ M), with 0 mM Mg²⁺ and 0.2 mM Pi present in the medium. The models were fitted to these data sets by setting $\Delta\Psi$ at 140 mV for the data corresponding to 0 mM Pi (A) and 190 mV for the data corresponding to ≥ 1 mM Pi (B–E). Matrix [Ca²⁺]_i was fixed at 250 nM and matrix [Mg²⁺]_i was fixed at 0.4 mM.

physiological variations in matrix [Ca²⁺]_i, which is demonstrated in Fig. S3, A and B, in Appendix SC in the [Supporting Material](#). Hence, the parameter estimates would depend neither on matrix [Ca²⁺]_i, nor on Ca²⁺ buffering and Ca²⁺-Pi complex formation in the matrix.

One of the key observations from the model analyses of the data (fitting) is that, the fitting is able to provide unique

and accurate estimates of the kinetic parameters K_C^0 , K_M^0 , K_{Pi} , and γ for all the available data sets (Table 1), irrespective of the amounts of Mg²⁺ and Pi present in the buffer. Furthermore, the fitting gives a unique and accurate estimate of k^0 for all the data sets obtained from the same source (e.g., k^0 is same for all the data curves of Crompton et al.). The fitting is able to provide a unique and accurate



2.0 μ M). Model simulations were done using $\Delta\Psi = 190$ mV, matrix [Ca²⁺]_i = 0.1 μ M, and matrix [Mg²⁺]_i = 0.4 mM. The rate constant k^0 was found to be 5.96 nM/s (0.006×10^{-3} nmol/mg/s), as 1 mg mitochondrial protein = 1 μ L mitochondrial volume.

FIGURE 3 Comparison of the Ca²⁺ uniporter model (lines) to the experimental data (points) on the cytosolic Ca²⁺ and Mg²⁺ dependencies of mitochondrial Ca²⁺ uptake measured in permeabilized cells by Szanda et al. (8). Shown are the simulations and comparisons of Model 2 of the uniporter to the kinetic data in which the initial rates of Ca²⁺ uptake were measured (A) as a function of cytosolic [Ca²⁺]_i (logarithmic y axis) with four different levels of cytosolic Mg²⁺ ([Mg²⁺]_e = 0.25, 0.5, 1, and 2 mM), and (B) as a function of cytosolic [Mg²⁺]_i (linear y axis) with three different levels of cytosolic Ca²⁺ ([Ca²⁺]_e = 0.5, 1.0, and 2.0 μ M). Model simulations were done using $\Delta\Psi = 190$ mV, matrix [Ca²⁺]_i = 0.1 μ M, and matrix [Mg²⁺]_i = 0.4 mM. The rate constant k^0 was found to be 5.96 nM/s (0.006×10^{-3} nmol/mg/s), as 1 mg mitochondrial protein = 1 μ L mitochondrial volume.

TABLE 1 Estimated parameter values and the corresponding normalized sensitivity coefficients for the kinetic model of mitochondrial Ca^{2+} uniporter

Parameters(ϕ_i)	Model 1		Model 2		Model 3		References
	Value	$S_{\phi_i}^E$	Value	$S_{\phi_i}^E$	Value	$S_{\phi_i}^E$	
k^0 (nmol/mg/s)	1.27×10^{-3}	0.0671	1.49×10^{-3}	0.2692	1.58×10^{-3}	0.3302	(9)
	20.32×10^{-3}	0.0671	22.35×10^{-3}	0.2692	22.12×10^{-3}	0.3302	(11)
	16.51×10^{-3}	0.0671	19.37×10^{-3}	0.2692	20.54×10^{-3}	0.3302	(14)
	27.94×10^{-3}	0.0671	30.55×10^{-3}	0.2692	29.23×10^{-3}	0.3302	(15)
	63.5×10^{-3}	0.0671	62.58×10^{-3}	0.2692	60.04×10^{-3}	0.3302	(16)
K_C^0 (M)	3.965×10^{-6}	0.1131	2.786×10^{-6}	0.4946	1.955×10^{-6}	0.7319	All
K_M^0 (M)	0.655×10^{-3}	0.0564	0.542×10^{-3}	0.1725	0.403×10^{-3}	0.4218	All
γ (Unitless)	3.9	0.1784	5.7	0.1739	9.2	0.1412	All
K_{Pi} (M)	0.2×10^{-3}	0.0319	0.2×10^{-3}	0.1678	0.2×10^{-3}	0.2696	All
E_ϕ ; S_ϕ^E	0.077; 0.447		0.058; 1.278		0.053; 1.895		Eqs. 10 and 11

Rate constant k^0 is redefined as $k^0 = [\text{T}]_{\text{tot}} \cdot k^0$. Note that the apparent values of K_C^0 and K_M^0 calculated using Eq. 9 for different $[\text{Pi}]$ matches well with the estimates in Table S1 of Appendix SB in the Supporting Material, obtained without considering the Pi regulation mechanism of K_C^0 and K_M^0 .

estimate of the parameter K_{Pi} ($K_{\text{Pi}} \approx 0.2$ mM) that governs the Pi-dependencies of the kinetic parameters (K_C^0 and K_M^0). Most importantly, Pi only seems to modulate the K_C^0 and K_M^0 in low Pi range (0–1 mM); for $[\text{Pi}]_e > 1$ mM; it does not alter K_C^0 and K_M^0 significantly, which is consistent with the available data.

Another important observation from the fitting is that this estimated value of K_C^0 is found to be much lower (e.g., $K_C^0 \approx 4.14$ μM using Model 1 for all data sets) compared to our previously estimated values (20,22) (e.g., $K_C^0 = 46$ μM and 88 μM using Model 1 from the data sets of Scarpa and colleagues (14,15)). This is due to the presence of 2 and 5 mM Mg^{2+} and 5 and 2 mM Pi in their buffer, which significantly modifies the apparent K_m of Ca^{2+} for the uniporter; this was not accounted for in our previous uniporter models (20,22). Even in the presence of 2 or 5 mM Pi, the apparent K_m of Ca^{2+} becomes ~ 9 μM , which is still much lower than that of the previous estimates. A small K_m value indicates that Ca^{2+} uptake attains saturation for a lower level of extramatrix Ca^{2+} , compared to our previous uniporter models (20,22), which is demonstrated in Fig. S4, A and B, of Appendix SC in the Supporting Material. Overall, this uniporter model accurately depicts the kinetics of mitochondrial Ca^{2+} uptake over a wide range of experimental conditions, in which the kinetics of Ca^{2+} uptake is hyperbolic in the absence of Mg^{2+} , sigmoid in presence of Mg^{2+} , and the degree of sigmoidicity increases with increase in $[\text{Mg}^{2+}]$, which is depicted in Fig. S4, C and D, of Appendix SC in the Supporting Material.

From the fittings of the models to the data (Fig. 2), it is apparent that both Model 2 and Model 3 adequately explain the available data compared to Model 1. This is also evident from the combined least-square error (E_ϕ) and normalized sensitivity coefficients for individual parameters and all parameters together ($S_{\phi_i}^E$ and S_ϕ^E) (Table 1). Specifically, E_ϕ is higher and S_ϕ^E is lower for Model 1, whereas E_ϕ is lower and S_ϕ^E is higher for Model 2 and Model 3. Additional statistical test based on the Akaike's Information Criterion

(AIC) is also conducted to verify the goodness of the fits. The computed value of AIC ($\text{AIC} = N_{\text{data}} \ln(E_\phi) + 2N_\phi$) was found to be higher for Model 1 compared to Model 2 and Model 3. This is obvious because the number of unknown parameters (N_ϕ) is the same in all three models. Therefore, AIC is strictly dependent on E_ϕ for a given model. From model parameterization and sensitivity analyses (Table 1) it is made clear that the kinetic parameters K_C^0 , K_M^0 , and γ are very accurate and consistent over a wide variety of experimental preparations at physiological levels of Pi. The estimated values of these parameters varied between different versions of the model due to the nature of binding assumptions. Therefore, either of the models (Model 2 or Model 3) can be considered as the most accurate model of the uniporter that best describes the available data compared to Model 1.

Based on these estimated values of the kinetic parameters (K_C^0 , K_M^0 , K_{Pi} , γ) and the previously estimated values of the biophysical parameters (α , nH), we further validated the uniporter model by simulating the extramatrix Ca^{2+} and Mg^{2+} dependencies of mitochondrial Ca^{2+} uptake in permeabilized cells (8). Fig. 3 A shows the model comparisons (logarithmic y axis) to the data as a function of cytosolic $[\text{Ca}^{2+}]$ for four different cytosolic $[\text{Mg}^{2+}]$ values ($[\text{Mg}^{2+}]_e = 0.25, 0.5, 1$, and 2 mM; $[\text{Pi}]_e = 1$ mM), whereas Fig. 3 B shows the model comparisons (linear y axis) to the data as a function of cytosolic $[\text{Mg}^{2+}]$ for three different cytosolic $[\text{Ca}^{2+}]$ values ($[\text{Ca}^{2+}]_e = 0.5, 1$, and 2 μM ; $[\text{Pi}]_e = 1$ mM). We used one of the most accurate models of the uniporter (Model 2) for these model simulations. The value of the rate constant k^0 was first converted from the units of nmol/mg/s to the units of nM/s by using the conversion factor 1 mg protein = 3.7 μL , then modified until the model predictions closely match the data. Matrix $[\text{Ca}^{2+}]$ was set at 250 nM, matrix $[\text{Mg}^{2+}]$ was set at 0.4 mM, and $\Delta\Psi$ was set at 190 mV. Considering that no other kinetic parameter was adjusted, the model is able to adequately predict the data and the nature of Mg^{2+} inhibition of mitochondrial Ca^{2+} uptake in intact permeabilized cells (8),

under physiological variations of cytosolic [Ca²⁺] and [Mg²⁺], supporting the accuracy of the model. Similar model predictions on % Mg²⁺ inhibition of Ca²⁺ uptake as a function of extramatrix [Ca²⁺] and [Mg²⁺] are demonstrated through Fig. S5 of Appendix SC in the [Supporting Material](#).

From the model analyses of the available data (Figs. 2 and 3), it is evident that this uniporter model is able to explain adequately the extramatrix Pi-, Mg²⁺-, Ca²⁺-, and ΔΨ-dependent kinetics of mitochondrial Ca²⁺ uptake in a wide variety of experimental preparations. To further validate the proposed mechanisms of Mg²⁺ inhibition of the uniporter function, we integrated this uniporter model and our recently developed 3Na⁺-1Ca²⁺ antiporter model (29) to our previously developed model of mitochondrial Ca²⁺ handling and bioenergetics (19). The integrated model is used to simulate the experimental data of Cox and Matlib (32) and McCormack et al. (12) on matrix free [Ca²⁺] as a function of extramatrix free [Ca²⁺], [Mg²⁺], and [Na⁺]. Detailed analyses and fittings of the integrated model to these data sets are described in Appendix SD (Figs. S6 and S7) in the [Supporting Material](#). The model appropriately described these data sets, further signifying the accuracy and robustness of this uniporter model and the proposed Mg²⁺ and Pi regulation mechanisms.

DISCUSSION

The work reported in this article provides the detailed development of a mechanistic mathematical model of mitochondrial Ca²⁺ uniporter that characterizes the kinetic mechanisms by which cytosolic Mg²⁺ inhibits and Pi regulates the uniporter-mediated Ca²⁺ transport into mitochondria. The model is extended from our previous models of the uniporter (20,22) and developed based on a similar multistate catalytic binding and interconversion mechanism for carrier-mediated facilitated transport and Eyring free energy barrier theory for interconversion and electrodiffusion (26–28). The model is parameterized and validated by comparing the model-simulated outputs to a large number of independent data sets on the kinetics of extramatrix Ca²⁺- and ΔΨ-dependent Ca²⁺ uptake via the uniporter, obtained in respiring mitochondria isolated from rat hearts and livers (9,11,14–16) and intact permeabilized myocytes (8) both in the absence and presence of different amounts of extramatrix Mg²⁺ and Pi. The model precisely describes the inhibitory effect of extramatrix Mg²⁺ and the regulatory role of Pi on the uniporter-mediated mitochondrial Ca²⁺ uptake, observed in these experiments.

Although this uniporter model has all the characteristics of our previous uniporter models (20,22), it significantly differs from the earlier models in that it mechanistically characterizes the inhibitory effect of Mg²⁺ and regulatory role of Pi on the uniporter function. It identifies the most plausible Mg²⁺ inhibition and Pi regulation mechanisms

based on a large number of experimental data sets (8,9,11,12,14–16). This uniporter model clarifies the earlier contradictory results on the type of Mg²⁺ inhibition of the uniporter function. Particularly, whereas the kinetic studies of Crompton et al. (9) and Scarpa and colleagues (14,15) showed a competitive type of Mg²⁺ inhibition, the later kinetic study of Bragadin et al. (11) showed a noncompetitive type of Mg²⁺ inhibition of Ca²⁺ uptake. Our model suggests a mixed-type mechanism for Mg²⁺ inhibition of Ca²⁺ uptake, in which the kinetics of Ca²⁺ uptake is hyperbolic in the absence of Mg²⁺ and sigmoid in the presence of Mg²⁺. This is manifested by a marked increase in the apparent K_m of Ca²⁺ for the uniporter and decrease in the maximal activity of the uniporter with increasing extramatrix [Mg²⁺]. Furthermore, the degree of sigmoidicity increases with increase in extramatrix [Mg²⁺], which is consistent with experimental observations.

This uniporter model also accurately explains the regulatory effect of cytosolic Pi on mitochondrial Ca²⁺ uptake, typically seen at low Pi levels (0–1 mM) in the kinetic study of Crompton et al. (9). Accordingly, the K_m of Ca²⁺ and Mg²⁺ for the uniporter (K_C^0 and K_M^0) were expressed in terms of Pi, in which Pi is shown to increase the value of K_C^0 and decrease the value of K_M^0 , and the Pi effect is shown to be negligible at higher Pi concentrations ([Pi] > 1 mM), consistent with the available data. On the other hand, without considering the Pi regulation mechanism, the model did not fit to the available data with unique K_C^0 and K_M^0 values for all data sets. This is apparent from the detailed model analyses presented in Appendix SB in the [Supporting Material](#), which justifies this formulation of Pi regulation of K_C^0 and K_M^0 parameters. However, additional kinetic data at submillimolar Pi concentrations may be necessary to mechanistically evaluate the precise regulatory effect of Pi on the uniporter function.

Despite significant technical advances, the structure and composition of mitochondrial Ca²⁺ uniporter and the regulation mechanisms of the uniporter function by divalent cations (e.g., Mg²⁺) and metabolites (e.g., Pi) are still largely unknown. Our present uniporter model is based on the assumption that there exist at least two binding sites for Ca²⁺ ions and two separate allosteric sites for Mg²⁺ ions on the uniporter (Fig. 1). We explored three possible Ca²⁺ and Mg²⁺ binding schemes (Model 1, full cooperativity; Model 2, partial cooperativity; and Model 3, no cooperativity) to resolve the most likely Ca²⁺ and Mg²⁺ binding mechanism that can explain the observed kinetics of mitochondrial Ca²⁺ uptake under diverse experimental preparations (8,9,11,12,14–16). With our model analyses of the available data, though we are not able to fully distinguish between the three versions of the kinetic model, Model 2 and Model 3 seem to describe the data most accurately compared to Model 1. Alternatively, by assuming 1), two binding sites for Ca²⁺ ions and one separate allosteric site for Mg²⁺ ions on the uniporter or 2), two binding sites for

Ca^{2+} ions and Mg^{2+} ions competing for the Ca^{2+} binding sites, we were not able to fit the two reduced models to any of the available data sets (see Figs. S8–S10 and Appendix SE in the Supporting Material).

In this mixed-type Mg^{2+} inhibition scheme for the Ca^{2+} uniporter, we consider binding of two Ca^{2+} ions and two Mg^{2+} ions to the uniporter, and the order of Ca^{2+} and Mg^{2+} binding does not affect the uniporter conformational states. In addition, the translocation of Ca^{2+} across the IMM occurs only via the conformational changes of the 2Ca^{2+} -bound uniporter complexes $\text{T}2\text{Ca}^{2+}_e$ and $2\text{Ca}^{2+}_x\text{T}$ (Fig. 1); other Mg^{2+} and Ca^{2+} -bound uniporter states do not undergo any conformational changes. However, alternate mechanisms are possible, in which the Mg^{2+} -bound uniporter states can undergo conformational changes, leading to Mg^{2+} influx via the Ca^{2+} uniporter. This is not considered here because it not only introduces additional parameter, but also there is lack of experimental evidence and data available on the kinetics of Mg^{2+} transport via the Ca^{2+} uniporter. Another salient feature of this uniporter model is that, using a single control parameter γ , the model can display all possible types of Mg^{2+} inhibition mechanisms (e.g., $\gamma = 1$, noncompetitive inhibition; $\gamma = \infty$, competitive inhibition; and $1 < \gamma < \infty$, a mixed-type inhibition). Consequently, by analyzing a large number of experimental data sets (8,9,11,12,14–16), the model confirmed that a mixed-type inhibition mechanism is most viable for the operation of the Ca^{2+} uniporter.

Another important feature of this uniporter model is that it provides a unique and consistent estimates of the kinetic parameters associated with the binding of Ca^{2+} , Mg^{2+} , and Pi to the uniporter (K^0_C , K^0_M , K_{Pi} , and γ) over a large number of experimental data sets (9,11,14–16). Most importantly, the estimate of K_m of Ca^{2+} for the uniporter (K^0_C) is found to be much lower compared to that reported in our earlier studies (20,22), and is unique over all the available data sets, irrespective of the amounts of extramatrix Mg^{2+} and Pi and the tissue types (hearts versus livers). For example, in this uniporter model, K^0_C is found to be $\sim 4.14 \mu\text{M}$ using Model 1 for all the available data sets, whereas Model 1 of our previous uniporter models (20,22) estimated a much higher value of K^0_C , at $\sim 44 \mu\text{M}$ for rat liver mitochondria (14) and $88 \mu\text{M}$ for rat heart mitochondria (15), due to the presence of different amount of Mg^{2+} and Pi in the buffer. In this model, this effect is attributed to changes in the binding affinities of Ca^{2+} for the uniporter with different amount of extramatrix Mg^{2+} and Pi. Accordingly, Ca^{2+} uptake reaches saturation at a much lower level of extramatrix Ca^{2+} ($\approx 30 \mu\text{M}$) in the absence of extramatrix Mg^{2+} compared to our previous uniporter models (20,22), when simulated over the entire $\Delta\Psi$ -range; in the presence of 2 mM Mg^{2+} , a higher level of extramatrix Ca^{2+} ($\approx 100 \mu\text{M}$) is required for the Ca^{2+} uptake to attain saturation (see Fig. S4, A and B, of Appendix SC in the Supporting Material).

Though the pathways of mitochondrial Ca^{2+} transport have been known for last three decades, their functional consequences are becoming increasingly clear in recent years in relation to the cell physiology and pathophysiology. Most importantly, mitochondrial Ca^{2+} uptake plays a substantial role in shaping cytosolic Ca^{2+} signals in many cell types and regulating local Ca^{2+} concentration in cellular microdomains (3,33,34). It is also apparent that mitochondria can considerably alter the kinetics of intracellular Ca^{2+} oscillations and waves generated by Ca^{2+} release from the endoplasmic reticulum (35). A decline in mitochondrial Ca^{2+} uptake can have significant effect on the spatiotemporal characteristics of cellular Ca^{2+} signaling and mitochondrial energy metabolism.

Considering its significance in tuning Ca^{2+} -mediated cellular responses, it is important to understand each factor that modulates mitochondrial Ca^{2+} transport. In this regard, Mg^{2+} is known to be a key regulator of mitochondrial cation and substrate transport systems, and inhibition of mitochondrial Ca^{2+} uptake by physiological elevations of cytosolic Mg^{2+} has been well recognized in many experimental studies (8,9,11). However, the consequences of this inhibition are poorly understood at cellular levels and it is largely unknown whether Mg^{2+} has any potential role in modifying cellular Ca^{2+} signals. Therefore, this model of Ca^{2+} uniporter is very useful for building biophysically based integrated models of mitochondrial and intracellular Ca^{2+} handling, to understand the mechanisms by which Ca^{2+} plays a critical role in the signal transduction processes and modulates mitochondrial and cellular energy metabolisms.

SUPPORTING MATERIAL

Five appendices, two tables, and 11 figures are available at [http://www.biophysj.org/biophysj/supplemental/S0006-3495\(11\)01119-2](http://www.biophysj.org/biophysj/supplemental/S0006-3495(11)01119-2).

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REFERENCES

1. Gunter, T. E., K. K. Gunter, ..., C. E. Gavin. 1994. Mitochondrial calcium transport: physiological and pathological relevance. *Am. J. Physiol.* 267:C313–C339.
2. Bernardi, P. 1999. Mitochondrial transport of cations: channels, exchangers, and permeability transition. *Physiol. Rev.* 79:1127–1155.
3. Duchen, M. R. 2000. Mitochondria and Ca^{2+} in cell physiology and pathophysiology. *Cell Calcium.* 28:339–348.
4. O'Rourke, B., S. Cortassa, and M. A. Aon. 2005. Mitochondrial ion channels: gatekeepers of life and death. *Physiology (Bethesda)*. 20:303–315.

5. Gunter, T. E., and S. S. Sheu. 2009. Characteristics and possible functions of mitochondrial Ca²⁺ transport mechanisms. *Biochim. Biophys. Acta.* 1787:1291–1308.
6. Romani, A., and A. Scarpa. 1990. Hormonal control of Mg²⁺ transport in the heart. *Nature.* 346:841–844.
7. Murphy, E., C. Steenbergen, ..., R. E. London. 1989. Cytosolic free magnesium levels in ischemic rat heart. *J. Biol. Chem.* 264:5622–5627.
8. Szanda, G., A. Rajki, ..., A. Spät. 2009. Effect of cytosolic Mg²⁺ on mitochondrial Ca²⁺ signaling. *Pflugers Arch.* 457:941–954.
9. Crompton, M., E. Sigel, ..., E. Carafoli. 1976. A kinetic study of the energy-linked influx of Ca²⁺ into heart mitochondria. *Eur. J. Biochem.* 69:429–434.
10. Crompton, M., M. Hediger, and E. Carafoli. 1978. The effect of inorganic phosphate on calcium influx into rat heart mitochondria. *Biochem. Biophys. Res. Commun.* 80:540–546.
11. Bragadin, M., T. Pozzan, and G. F. Azzone. 1979. Kinetics of Ca²⁺ carrier in rat liver mitochondria. *Biochemistry.* 18:5972–5978.
12. McCormack, J. G., H. M. Browne, and N. J. Dawes. 1989. Studies on mitochondrial Ca²⁺-transport and matrix Ca²⁺ using Fura-2-loaded rat heart mitochondria. *Biochim. Biophys. Acta.* 973:420–427.
13. Wan, B., K. F. LaNoue, ..., R. C. Scaduto, Jr. 1989. Regulation of citric acid cycle by calcium. *J. Biol. Chem.* 264:13430–13439.
14. Vinogradov, A., and A. Scarpa. 1973. The initial velocities of calcium uptake by rat liver mitochondria. *J. Biol. Chem.* 248:5527–5531.
15. Scarpa, A., and P. Graziotti. 1973. Mechanisms for intracellular calcium regulation in heart. I. Stopped-flow measurements of Ca²⁺ uptake by cardiac mitochondria. *J. Gen. Physiol.* 62:756–772.
16. Wingrove, D. E., J. M. Amatruda, and T. E. Gunter. 1984. Glucagon effects on the membrane potential and calcium uptake rate of rat liver mitochondria. *J. Biol. Chem.* 259:9390–9394.
17. Magnus, G., and J. Keizer. 1997. Minimal model of β -cell mitochondrial Ca²⁺ handling. *Am. J. Physiol.* 273:C717–C733.
18. Nguyen, M. H., S. J. Dudycha, and M. S. Jafri. 2007. The effects of Ca²⁺ on cardiac mitochondrial energy production is modulated by Na⁺ and H⁺ dynamics. *Am. J. Physiol. Cell Physiol.* 292:2004–2020.
19. Dash, R. K., and D. A. Beard. 2008. Analysis of cardiac mitochondrial Na⁺-Ca²⁺ exchanger kinetics with a biophysical model of mitochondrial Ca²⁺ handling suggests a 3:1 stoichiometry. *J. Physiol.* 586:3267–3285.
20. Dash, R. K., F. Qi, and D. A. Beard. 2009. A biophysically based mathematical model for the kinetics of mitochondrial calcium uniporter. *Biophys. J.* 96:1318–1332.
21. Bazil, J. N., G. T. Buzzard, and A. E. Rundell. 2010. Modeling mitochondrial bioenergetics with integrated volume dynamics. *PLOS Comput. Biol.* 6:e1000632.
22. Pradhan, R. K., F. Qi, ..., R. K. Dash. 2010. Characterization of membrane potential dependency of mitochondrial Ca²⁺ uptake by an improved biophysical model of mitochondrial Ca²⁺ uniporter. *PLoS ONE.* 5:e13278.
23. Bose, S., S. French, ..., R. S. Balaban. 2003. Metabolic network control of oxidative phosphorylation: multiple roles of inorganic phosphate. *J. Biol. Chem.* 278:39155–39165.
24. Wu, F., F. Yang, ..., D. A. Beard. 2007. Computer modeling of mitochondrial tricarboxylic acid cycle, oxidative phosphorylation, metabolite transport, and electrophysiology. *J. Biol. Chem.* 282:24525–24537.
25. Nicholls, D. G. 2005. Mitochondria and calcium signaling. *Cell Calcium.* 38:311–317.
26. Läuger, P. 1973. Ion transport through pores: a rate-theory analysis. *Biochim. Biophys. Acta.* 311:423–441.
27. Läuger, P., and B. Neumcke. 1973. Theoretical analysis of ion conductance in lipid bilayer membranes. *Membranes.* 2:1–59.
28. Keener, J. P., and J. Sneyd. 1998. *Mathematical Physiology.* Springer, New York.
29. Pradhan, R. K., D. A. Beard, and R. K. Dash. 2010. A biophysically based mathematical model for the kinetics of mitochondrial Na⁺-Ca²⁺ antiporter. *Biophys. J.* 98:218–230.
30. Beard, D. A. 2005. A biophysical model of the mitochondrial respiratory system and oxidative phosphorylation. *PLOS Comput. Biol.* 1:e36.
31. Jung, D. W., E. Panzeter, ..., G. P. Brierley. 1997. On the relationship between matrix free Mg²⁺ concentration and total Mg²⁺ in heart mitochondria. *Biochim. Biophys. Acta.* 1320:310–320.
32. Cox, D. A., and M. A. Matlib. 1993. A role for the mitochondrial Na⁺-Ca²⁺ exchanger in the regulation of oxidative phosphorylation in isolated heart mitochondria. *J. Biol. Chem.* 268:938–947.
33. Brini, M. 2003. Ca²⁺ signaling in mitochondria: mechanism and role in physiology and pathology. *Cell Calcium.* 34:399–405.
34. Bers, D. M. 2008. Calcium cycling and signaling in cardiac myocytes. *Annu. Rev. Physiol.* 70:23–49.
35. Rizzuto, R., S. Marchi, ..., P. Pinton. 2009. Ca²⁺ transfer from the ER to mitochondria: when, how and why. *Biochim. Biophys. Acta.* 1787:1342–1351.